

How Far Off the Recorded Path Can a Gaussian-Splat Driving Simulator Be Trusted?

CS 566 Project Proposal · Kyle Sha and Allen Xia · September 29, 2026

Main papers: [1] **Difix3D+** (CVPR 2025 Oral), *the method we reimplement and retrain* · [2] **FaithFusion** (CVPR 2026), *faithfulness-aware progressive update* · [3] **HUGSIM** (TPAMI 2026), *simulator and closed-loop test*

1. Problem: what we are trying to solve

Driving stacks are increasingly tested in simulators built from real drives: the recorded video is reconstructed as a Gaussian splat, and the simulated car can then drive *differently* from the log (HUGSIM [3]). This only works if **renders from off-log viewpoints are real enough that the stack behaves as it would in reality**, yet splats degrade as the camera leaves the recorded path.

The current fix is a generative "**fixer**": **Difix3D+** [1] cleans off-path renders with a single-step diffusion model and distills them back into the splat; NVIDIA's driving-trained **Fixer** [4] uses the same recipe; **FaithFusion** [2] limits "over-restoration" with a per-pixel information-gain map. Our pilot shows such fixers **make renders prettier while inventing lane lines, road texture and vehicles**.

We will reimplement Difix3D+ ourselves, train it on driving data, and measure how trust decays along two deviation axes, lateral offset and camera height, against real off-path images, comparing fixers head to head on image quality, hallucination and perception agreement, plus whether trained fixers transfer to new cities. **Input:** recorded driving logs with poses. **Output:** street splats, trained fixers, and trust-vs-deviation curves.

2. Motivation: why we want to solve it

Splat simulation is becoming standard for testing driving stacks, but current fixes don't solve large view changes (EUVS [7]). The closest work leaves our question open: PRG-Fusion [6] shifts Waymo scenes 3 m and 6 m and ranks fixers, but with no real off-path images or hallucination accounting; a 2026 study [10] measures perception trust vs. offset without fixers; SetDiff [11] uses Para-Lane with image metrics only; MV2 [9] provides real cross-vehicle views but tests no fixers. **Our differentiator: real off-path ground truth, fixers compared head to head, explicit hallucination accounting ("prettier vs. more trustworthy"), and cross-dataset transfer.** Course material: camera models, homographies, multi-view geometry, radiometry, CNNs, 3D reconstruction.

3. Possible steps to the solution

1. **Street splats** (3DGS [14], gsplat) for Para-Lane [8] (the same road scanned in 3 lanes, about 3.3 m / 6.3 m apart), MV2 [9] (car and scooter in different lanes at different camera heights), EUVS [7] and HUGSIM [3] scenes.
2. **Training pairs** (Difix3D+ curation [1]; cf. ReCamDriving [12]): one lane's splat rendered at a parallel lane's poses vs. that lane's real images, plus cycle-reconstruction, under-fit and on-path identity pairs.
3. **Reimplement Difix3D+ [1]** on pretrained SD-Turbo (as the authors did): our own training code (single step at $\tau = 200$, reference-view mixing, LoRA decoder, L2 + LPIPS + Gram), **validated against NVIDIA's official trainer and released weights** (on DL3DV if access is granted, otherwise on our driving data at a matched budget), then trained on driving pairs; plus its **progressive 3D update**.
4. **Other arms:** released Difix, NVIDIA Fixer [4], FaithFusion's [2] information-gain weighting (and whether its map, or a no-reference reliability score as in ClearGS [13], predicts hallucinated pixels), **a classical road-surface prior** (height + slope field from LiDAR, ViSE-style [16]: no generation, so no hallucination), and **a small from-scratch U-Net** (the smaller-version ablation).
5. **Trust vs. deviation:**
 - **Lateral offset:** Para-Lane real adjacent-lane frames.
 - **Camera height:** MV2 car ↔ scooter (real ground truth, mixed with lateral offset) and a road-plane-warped reference on Para-Lane.
 - **Heading:** Para-Lane rotation-warped reference, limited to regions the original camera saw (newly revealed regions are reported separately).
 - **Metrics:** PSNR / SSIM / LPIPS; hallucination (lane-paint precision *and* recall, CLRNet [15] lane agreement, fake vehicles); NTA / NTL-IoU [5]; the HUGSIM closed loop with the released LTF policy, with one HUGSIM score reproduced vs. its paper.
6. **Transfer:** do fixers trained on Para-Lane help on EUVS (MARS) and MV2 (different countries and cameras)?

Learning components — what we do with each:

Component	In the paper	Our version
Difix fixer [1]	SD-Turbo fine-tuned with their recipe; official code and weights released	Reimplemented and retrained by us , validated against the official trainer and weights; fits the 3090 (about 16 GB)
Difix3D+ progressive update [1]	Per-scene optimization with pseudo-views	Reimplemented
FaithFusion weighting [2]	Information-gain map + restorer (weights not released)	Reproduced with their code , Difix as the restorer
NVIDIA Fixer [4]	Cosmos 0.6B trained on NVIDIA AV data	Frozen baseline (a 24 GB fine-tune fits)
Our U-Net (new)	—	Smaller version, trained from scratch (~15M parameters)
Road prior [16]; splats; CLRNet / YOLO / LTF	Road SDF; per-scene optimization; pretrained	Reimplemented; full optimization; frozen

Datasets: Para-Lane [8] (training + evaluation; CC BY-NC; single scenes 2–4 GB), MV2 [9] (evaluation; 12 GB; academic use), EUVS [7] (evaluation; Hugging Face terms), HUGSIM scenes [3] (NTA-IoU and closed loop; MIT). **Compute:** one RTX 3090; a scene splat about 5 min, Difix training about 16 GB, U-Net about 1 day.

Status (Sep 28 pilot): the full pipeline runs. Trust falls steeply with lateral offset and camera height (heading matters much less); zero-shot fixers improve LPIPS but not faithfulness; our driving-trained Difix is the most faithful but invents more vehicles and does not transfer to EUVS; the progressive update helps strongly on one test scene and little on the other; the road prior helps with no hallucination.

4. Rough timetable

Dates	Kyle	Allen
Sep 29 – Oct 12	MV2 splats and the height axis; more Para-Lane test scenes	Metric harness for MV2 / EUVS; per-scene variance analysis
Oct 13 – 29	Difix reimplement validation; more seeds	Road prior on all scenes; hallucination-map analysis. Mid-term report (Oct 29)
Oct 30 – Nov 20	Transfer study (EUVS, MV2); progressive update across scenes	HUGSIM closed loop (more seeds and fixers)
Nov 21 – Dec 10	Final curves; side-by-side videos	Presentation (Dec 1–10) and webpage (Dec 10)

Risks: DL3DV access (manual approval) may not arrive, in which case validation is on our data only. EUVS's rotation-only level needs nuPlan images (separate licence), so heading uses Para-Lane. Progressive gains are scene-dependent, so we report per-scene variance. UniAD / VAD need a legacy CUDA stack, so the closed loop uses LTF. Licences (NVIDIA weights, CC BY-NC, academic-use data): coursework only, no redistribution.

References (paper · code)

Main papers

1. Difix3D+: 3D reconstruction with single-step diffusion, CVPR 2025 (Oral) · arxiv.org/abs/2503.01774 · github.com/nv-tlabs/Difix3D
2. FaithFusion: pixel-wise information gain for reconstruction + generation, CVPR 2026 · arxiv.org/abs/2511.21113 · github.com/wangyuanbiubiubiu/FaithFusion
3. HUGSIM: closed-loop photorealistic AD simulator, TPAMI 2026 · arxiv.org/abs/2412.01718 · github.com/hyzhou404/HUGSIM

Supplementary

4. NVIDIA Fixer (Difix3D+ recipe, Cosmos 0.6B, AV data), 2025 · github.com/nv-tlabs/Fixer
5. ReconDreamer (NTA / NTL-IoU), CVPR 2025 · arxiv.org/abs/2411.19548
6. PRG-Fusion: generative priors routed by reconstruction evidence, 2026 · arxiv.org/abs/2609.06948
7. EUVS benchmark, ICCV 2025 · arxiv.org/abs/2412.05256 · github.com/ai4ce/EUVS-Benchmark
8. Para-Lane dataset, 3DV 2025 · arxiv.org/abs/2502.15635
9. MV2: multi-view multi-vehicle driving NVS dataset, ECCV 2026 · arxiv.org/abs/2608.12442 · huggingface.co/datasets/sanjay810/mv2-dataset
10. From Concept to Capability: evaluating 3DGS for AD, SafeComp 2026 · arxiv.org/abs/2605.01995
11. SetDiff: geometry-grounded set diffusion for NVS, 2026 · arxiv.org/abs/2601.07540
12. ReCamDriving: camera-controlled synthesis for novel trajectories, ECCV 2026 · arxiv.org/abs/2512.03621
13. ClearGS: reliability-aware 3DGS from handheld videos, 2026 · arxiv.org/abs/2609.31509
14. 3D Gaussian Splatting, SIGGRAPH 2023 · arxiv.org/abs/2308.04079
15. CLRNet (lane detector), CVPR 2022 · arxiv.org/abs/2203.10350 · github.com/Turoad/CLRNet
16. ViSE (road-surface prior), ICCV 2025 RealADSim winner · arxiv.org/abs/2510.18341